Physics - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Red light, obtained from an incandescent lamp by means of a conventional absorptive filter, may havewavetrainsonly about 10 \{}lambda long. If \{}lambda = 650...

   (a) 55 nm, 3.82 × 10ˆ13 Hz

   (b) 70 nm, 5.12 × 10ˆ13 Hz

   (c) 75 nm, 5.62 × 10ˆ13 Hz

   (d) 90 nm, 7.62 × 10ˆ13 Hz

   (e) 80 nm, 6.62 × 10ˆ13 Hz

2. The image produced by a concave mirror is at -16.2m, and the magnification is 1.79. What is the object distance in terms of meter?

   (a) 6.34

   (b) 9.05

   (c) 14.7

   (d) 8.46

   (e) 11.2

3. Two homogeneous cylinders made of the same material rotate about their own axes at the same angular velocity \{}cyrchar\{}cyromega. What is the ratio between...

   (a) 1:8

   (b) 1:2

   (c) 16:1

   (d) 4:1

   (e) 1:1

4. A certain element emits 1 alpha particle, and its products then emit 2 beta particles in succession. The atomic number of the resulting element is changed by

   (a) minus 3

   (b) minus 2

   (c) minus 1

   (d) plus 1

   (e) plus 3

5. Suppose a student who was farsighted wears glasses that allows him to read at a distance of 20cm from his eyes to the book. His near-point distance is 63cm....

(a) 4.5
(b) 2.0
(c) 3.846
(d) 4.0
(e) 4.346

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Microscopic slush in water tends to make the water' is 'more slippery'.
2. True or False: The correct answer to 'The loudness of a sound is most closely related to its' is 'timbre'.
3. True or False: The correct answer to 'If the Sun collapsed to become a black hole Planet Earth would' is 'Both C and D.'.
4. True or False: The correct answer to 'A coil of 60 ohms inductive reactance in series with a resistance of 25 ohms is connected to an a-c line of 130 volts. What will be the...' is '2.5 amp'.
5. True or False: The correct answer to 'Two charged, massive particles are isolated from all influence except those between the particles. They have charge and mass such that...' is 'The particles must have the same sign of charge.'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. If you know both the actual brightness of an object and its apparent brightness from your location then with no other information you can estimate:
2. A small cart of mass m is initially at rest. It collides elastically with a large cart of mass 4m and velocity v. The large cart loses half its kinetic energy to the little cart. The little cart now has a velocity of

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